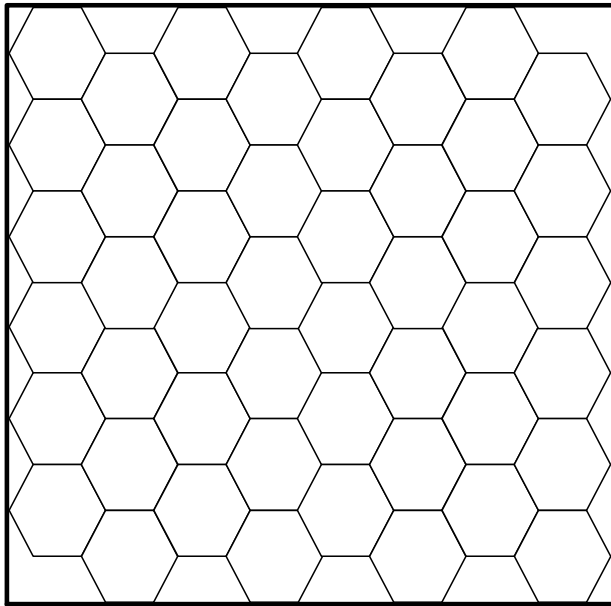
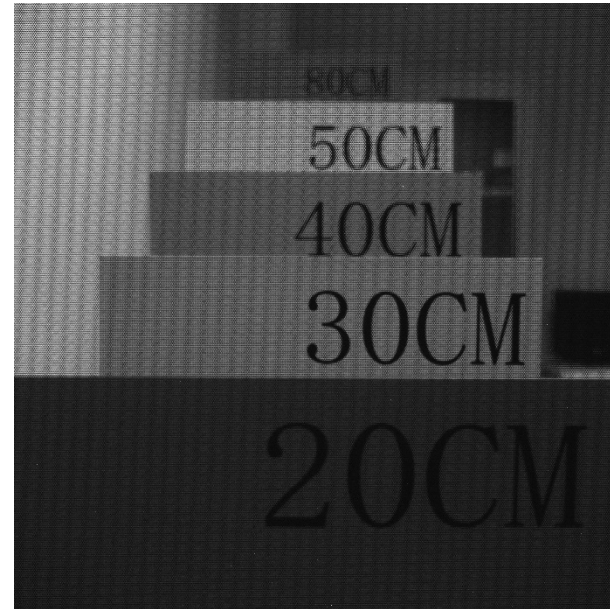


Part One

Process 1: Preprocessing



The Raw image is composed of hexagonal cells



Lytro_RAW

Export raw image to computer by using LYTRO Compatible Viewer.

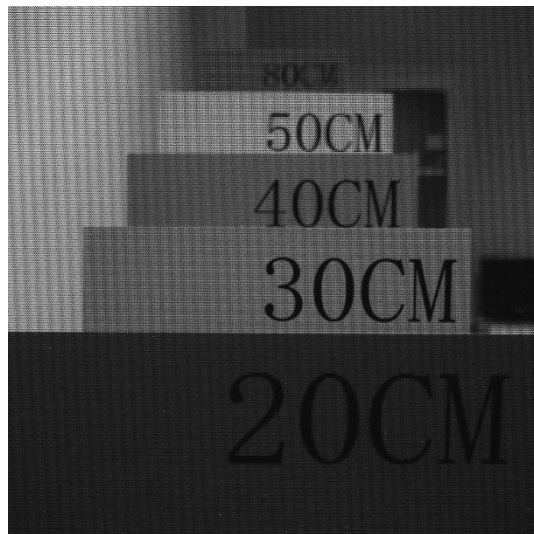
Part One

Process 1: Preprocessing

Demosaic Convert Bayer pattern encoded image to real color image

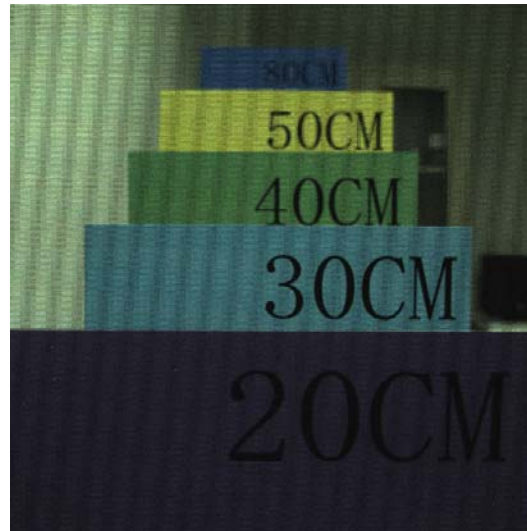
```
RGB = demosaic(I, sensorAlignment)
```

```
Lytro_RAW_Demosaic = im2double(demosaic(Lytro_RAW,'bggr'));
```



Lytro_RAW

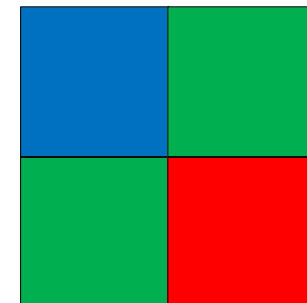
Demosaic →



Lytro_RAW_Demosaic

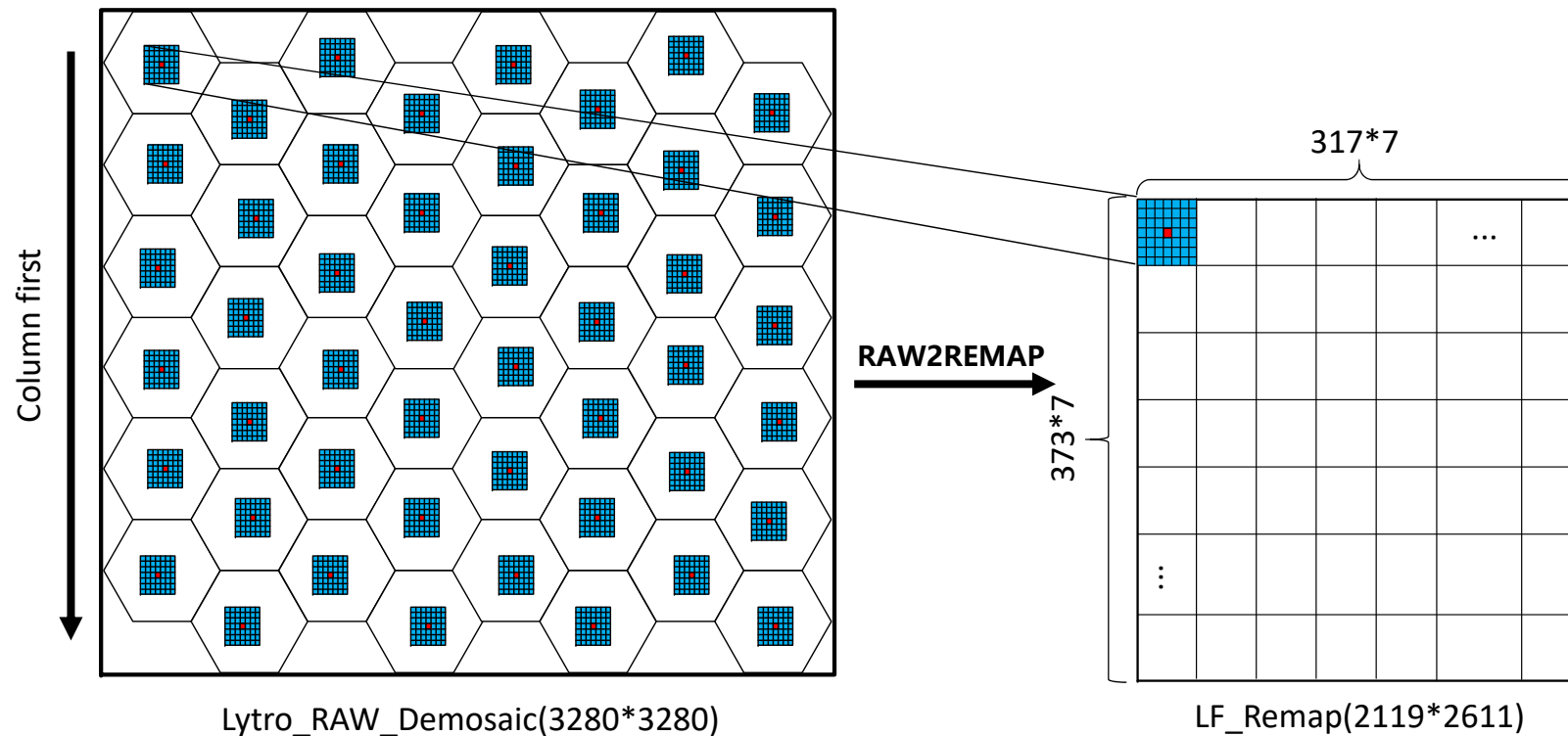
Blue	Green
Green	Red

sensorAlignment 'bggr'



Part One

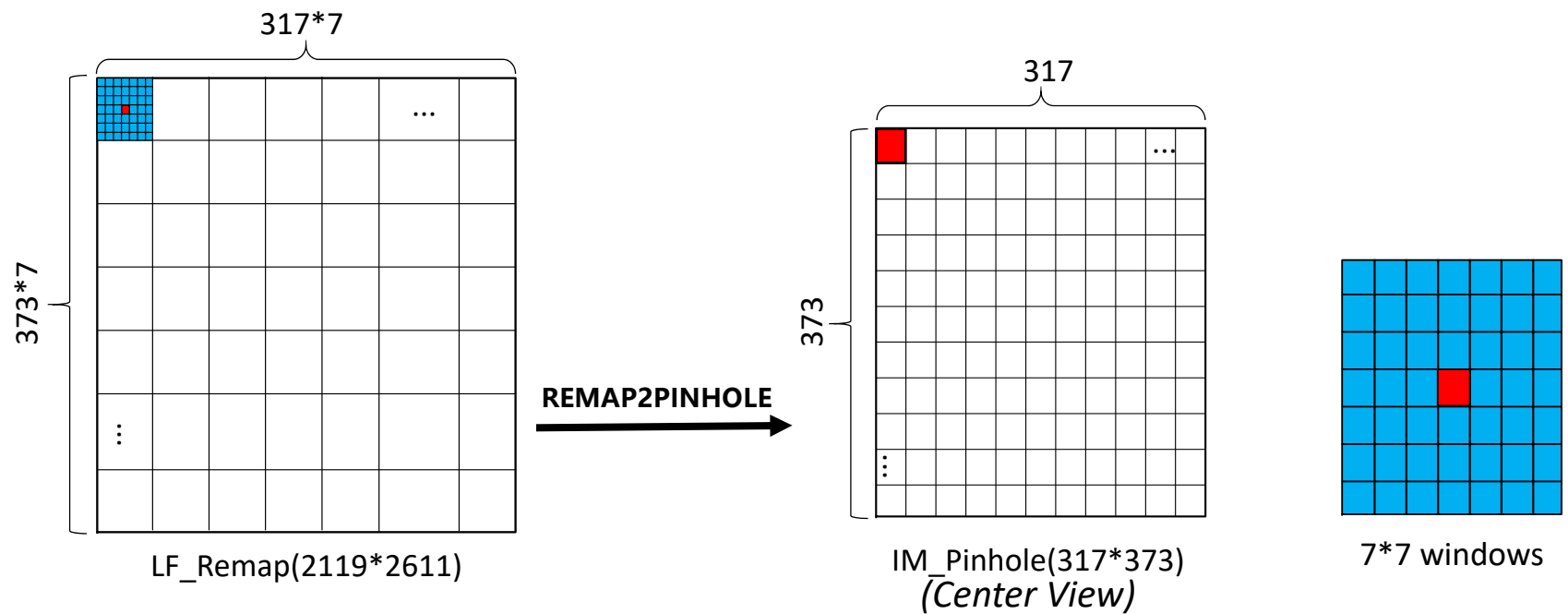
Process 1: Preprocessing



```
LF_Remap = RAW2REMAP(Lytro_RAW_Demosaic,image_cords_6,LF_parameters)
```

Part One

Process 1: Preprocessing

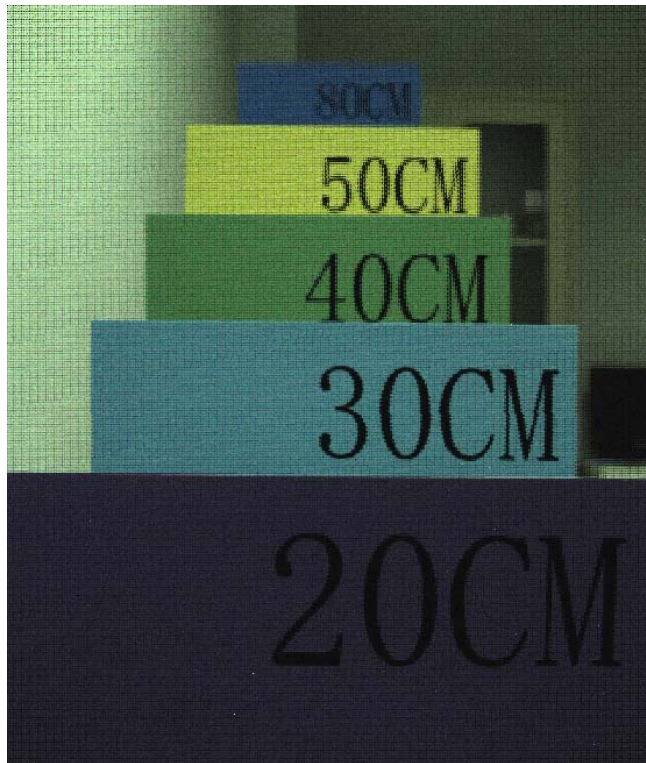


$IM_Pinhole = REMAP2PINHOLE(LF_Remap, LF_parameters)$

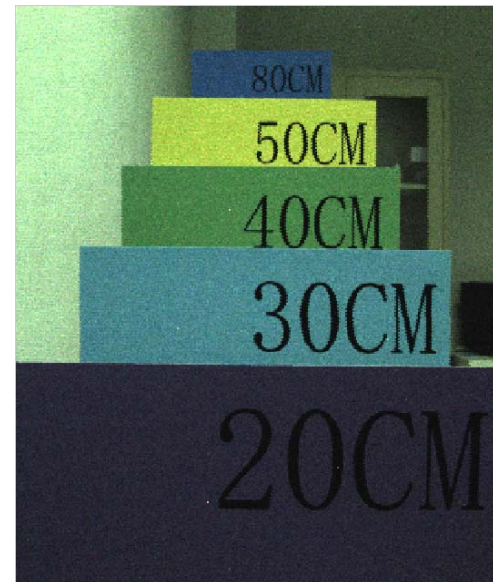
Part One

Process 1: Preprocessing

Result:



LF_Remap

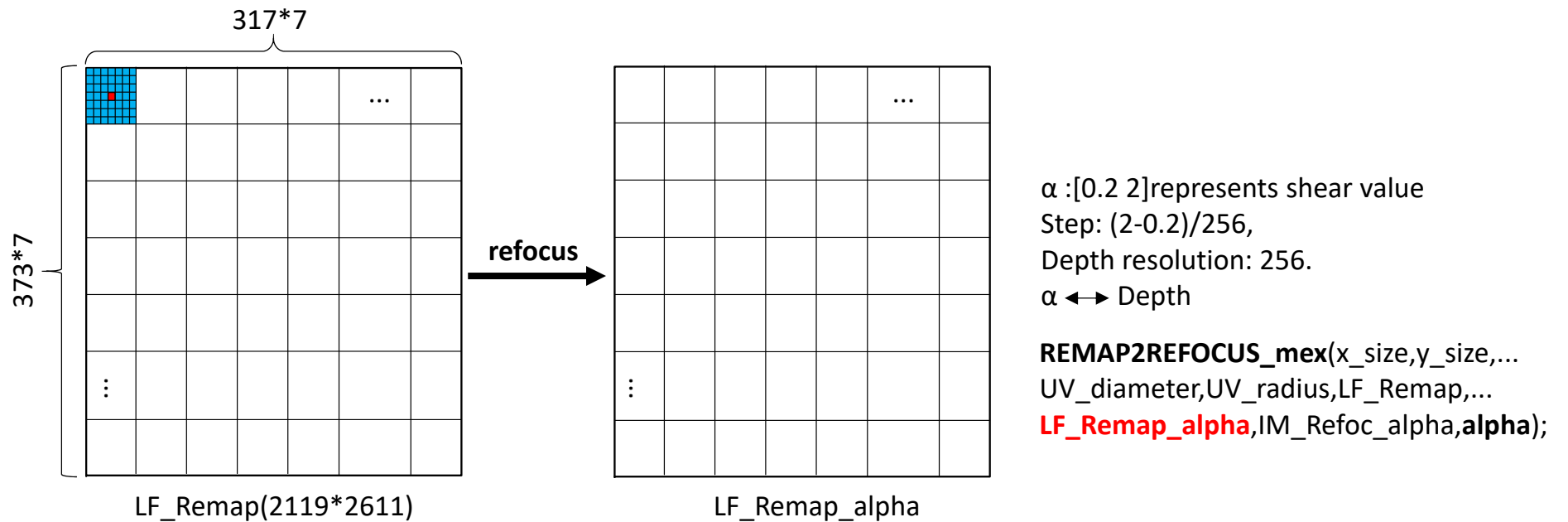


IM_Pinhole

Part One

Process 2: Refocusing

Refocus: shear EPI (referring to the paper: *Light Field Photography With A Hand-held Plenoptic Camera.*)



The interpretation of the formula

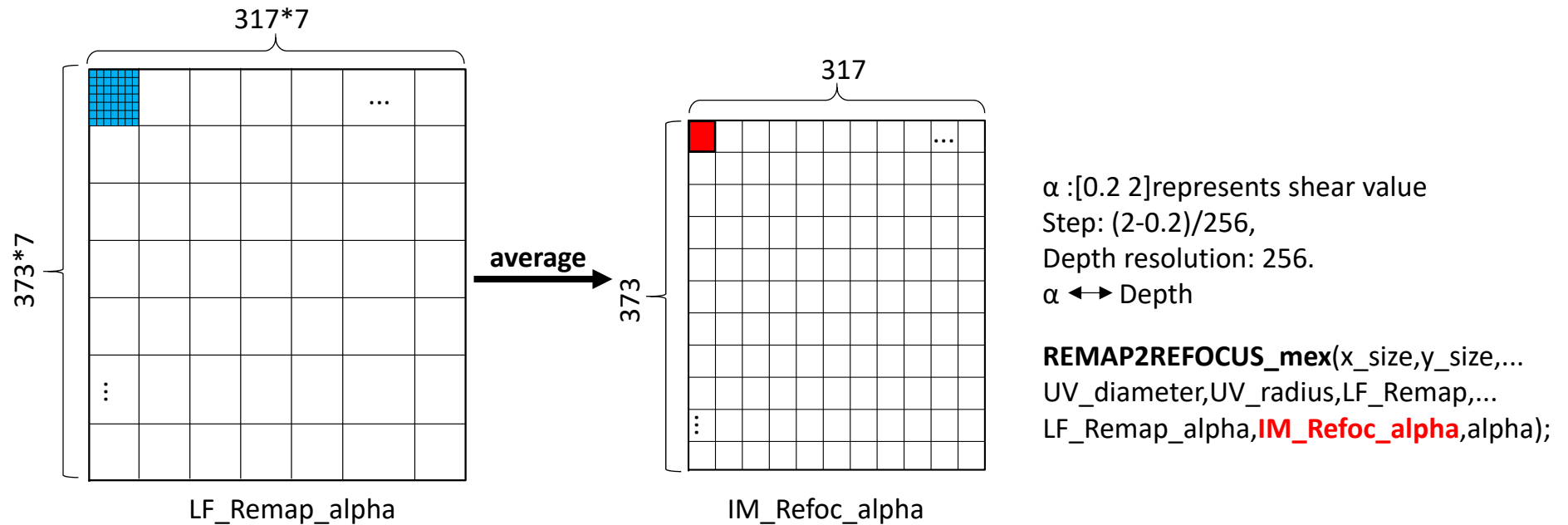
$$L_{\alpha}(x, y, u, v) = L_0 \left(x + u \left(1 - \frac{1}{\alpha} \right), y + v \left(1 - \frac{1}{\alpha} \right), u, v \right)$$

L_0 is the rectification of the captured image, or in our paper is **LF_Remap** image

L_{α} is the image at the depth α

Part One

Process 2: Refocusing

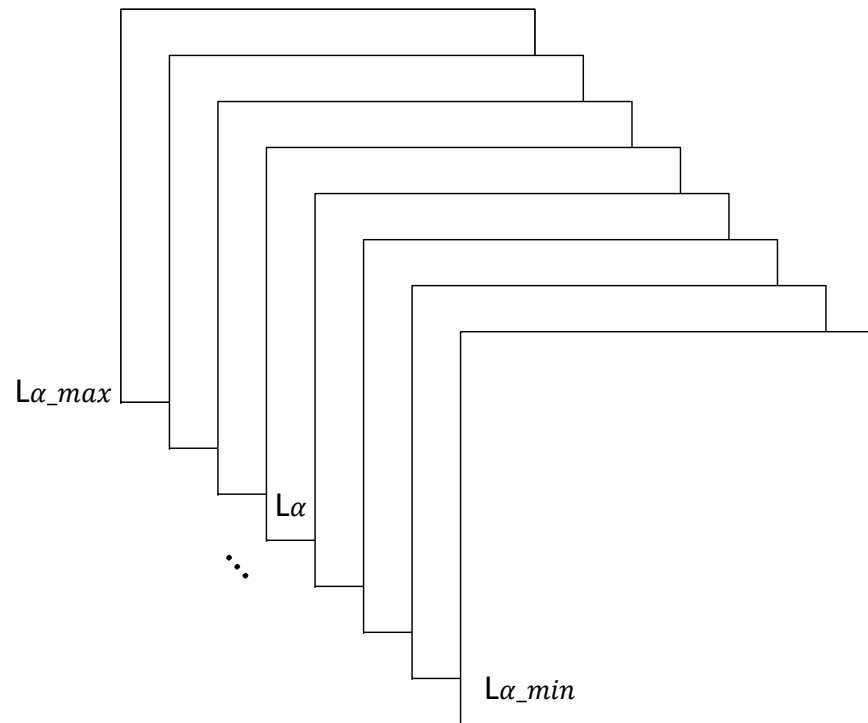


`output_image[y + x * height + 0 * height*width] = output_color_R/window_size;`

In order to get the intensity of RED in the pixel (x,y), we averaging the RED intensity in the window 7×7 in **LF_Remap_alpha**.

Part One

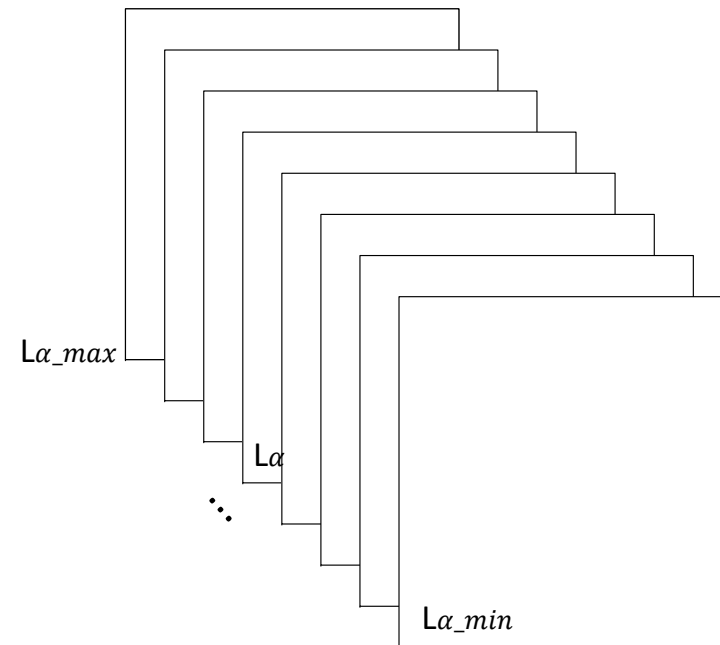
Process 2: Refocusing



LF_Remap_alpha



LF_Remap_alpha is used to calculate intensity range of the image.



IM_Refoc_alpha



temporary no use